

Urban Heat Island Mapping Using Multi-Seasonal Land Surface Temperature Modelling and SHAP-Based Interpretability

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Abstract—Urban heat islands in rapidly urbanising tropical cities are typically analysed through unified global models that fail to account for the structural differences between dense urban cores and rural peripheries. This paper presents a regime-stratified machine learning framework applied to Chennai and surrounding districts in Tamil Nadu, India, using Landsat-derived Land Surface Temperature (LST) data from summer 2023 and Post-Monsoon 2023. A dataset of 9,483 pixels was assembled across 12 features — including built-up density, NDVI, MNDWI, elevation, and ESA land use/land cover classes — and used to train Random Forest (RF) and Linear Regression (LR) models under urban and rural regime splits. Spatial block cross-validation was employed to prevent leakage from spatially autocorrelated pixels, yielding R^2 values ranging from 0.289 to 0.498 across 12 models. SHAP-based attribution revealed that NDVI is the dominant thermal regulator across all conditions, while MNDWI importance nearly doubles in urban zones during the post-monsoon period (0.254 to 0.622). In rural post-monsoon conditions, elevation unexpectedly surpasses vegetation indices as the leading feature (mean $|\text{SHAP}| = 0.651$), a physically interpretable result driven by terrain-induced cooling gradients. These findings establish that thermal drivers are structurally distinct across regimes and seasons, and that a global model obscures rather than reveals these mechanisms.

Index Terms— Urban Heat Island, Land Surface Temperature, SHAP, Random Forest, Chennai, Regime-Stratified Modelling, Remote Sensing

I. INTRODUCTION

The urban heat island (UHI) effect- which explains the tendency for built-up areas, cities to register a higher surface and air temperature than their surrounding rural land, has been documented across major cities across the entire continent. In South Asia, where urbanisation is advancing quite fast but infrastructure and green spaces aren't keeping up, infact they're seen to be declining rapidly. It is understood that research and development in spatial and seasonal heat patterns has direct implications for public health, energy demand, and urban planning.

A strong case for this kind of analysis would be Chennai. The city lies on a flat coastal plain in Tamil Nadu and has a tropical wet and dry climate, leading to clear seasonal differences in temperature. The Summers here are typically hot and dry with limited vegetation activity, while the post-monsoon period is noticeably cooler and wetter. Mean LST in the study dataset was 41.70°C in summer and 33.47°C in Post-Monsoon, a gap exceeding 8°C. This contrast between seasons gives us a kind of natural experiment. If the same set of factors control surface temperature in both seasons, then one model trained on all the data should work well overall. But here it doesn't, it means we need to treat each season separately and build more tailored models that would do justice to the study.

Current remote sensing studies of urban thermal environments for the most part, rely on global regression models or correlation frameworks applied uniformly across the study regions without accounting for diversity [1]–[3]. In doing so, although computationally efficient, these approaches tend to merge the thermal behaviour of dense urban cores with peri-urban agricultural and forested regions, which are dynamically different. Heat responses vary significantly between a built-up industrial pixel in Ambattur and an agricultural pixel in Kancheepuram, although global models would have assigned them equal weight in parameter estimation.

This paper makes three primary contributions. Firstly, it develops a framework that is regime-stratified which partitions pixels into two classes- urban and rural based on a built-up density threshold set ($\text{BuiltUp} \geq 4,976 \text{ m}^2$) and follows up by training independent models within each class. Second, To address this, spatial block cross-validation is implemented, ensuring that pixels that are geographically close do not appear in both training and testing sets, because that would wrongly inflate the metrics during performance. Third, SHAP (SHapley Additive exPlanations) [4] is used to break down feature contributions across regimes and seasons, showing exactly how the features act and change under different conditions. Which even after progression in interpretability techniques, seems to be missing in most UHI studies.

II. LITERATURE REVIEW

A. UHI Dynamics in Tropical and South Asian Cities

Urban heat island research in tropical cities has grown to expand considerably over the past two decades, driven by heightening concern over heat-related health impacts. Studies in Indian cities, including Delhi, Bengaluru, and Hyderabad, have documented UHI intensities to range from 1°C to 4°C depending on season and measurement method [1]. The postmonsoon period is relatively understudied compared to summer, despite being a distinct thermal phase where moisture plays a key role alongside built-up density.

B. Remote Sensing Approaches to LST Modelling

At a 30-metre resolution, Landsat thermal infrared data has become a common choice for pixel-level LST estimation, with indices such as NDVI and MNDWI serving as proxies in modelling [2]. Several studies have established a negative NDVI–LST relationships in urban areas, with NDVI explaining 30–60% of LST variance in most tropical city datasets. However, these relationships are usually based on global regression models rather than spatially stratified

approaches, leaving behind regime-level variation completely unexplored [3].

C. Machine Learning for Urban Thermal Prediction

Random Forest is widely used in LST prediction tasks because it handles non-linear relationships effectively without requiring assumptions about data distribution [5]. Studies applying RF to urban thermal modelling generally report R^2 values in the 0.40–0.65 range under standard random cross-validation, with performance seen to be declining under stricter spatial validation protocols [6]. The difference between random and spatial cross-validation performance is considered to be a key methodological issue, as random CV tends to overestimate accuracy by treating the nearby pixels as entirely independent observations [7].

D. SHAP for Geospatial Interpretability

SHAP-based interpretability’s application has increased in geospatial machine learning tasks such as air quality prediction, land cover classification, and flood susceptibility mapping [4]. In the UHI context, SHAP allows researchers to move beyond just aggregation of feature importance to specifically understanding each and every decision, direction, magnitude, and interaction of feature effects at the pixel level. This is particularly important when we are working in regime-based analyses, where a single feature may tend to contribute to cooling in one setting but lead to heating in another.

E. Research Gaps and Motivation

Despite the growing volume of research in this field, most LST studies are yet to separate data by urban and rural regimes, they do not apply spatial cross-validation, or use SHAP for deeper interpretation. Because of these gaps, many models struggle with how well models are supposed to generalize and how easily they can be physically interpreted. This study is designed to tackle all three of these gaps.

III. RESEARCH METHODOLOGY

A. Study Area

The study is focused on Chennai and its surrounding districts in Tamil Nadu, India (12.98°N–13.13°N, 80.05°E–80.23°E). The region is relatively compact but it’s a well-diversified mix of dense urban areas, clumped-up industrial plots, peri-urban expansion zones, agricultural land, sparsely seen vegetation, and water bodies. This level of diversity is what makes the region well suited for regime-based thermal analysis, as both the urban and rural extremes are captured within a relatively small area.

The Spatial patterns of Summer 2023 LST demonstrate a clear gradient. High-temperature zones ($>45^\circ\text{C}$) are concentrated in the northern and northwestern regions, corresponding to dense built-up and industrial areas. In contrast, relatively cooler regions ($35\text{--}38^\circ\text{C}$) are observed in the southeastern zones, where vegetation density is higher

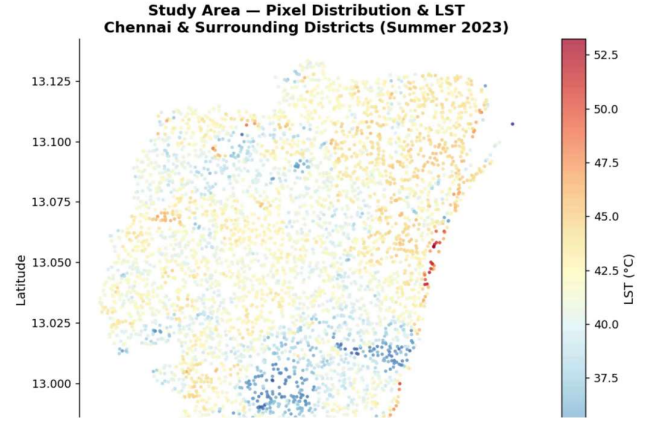


Fig. 1. Study Area’s Pixel Distribution & LST

B. Dataset Construction

Two pixel-level datasets were initially extracted from Google Earth Engine: Summer 2023 and Post-Monsoon 2023, containing 5,000 observations each. After applying preprocessing techniques, a total of 9,483 valid samples were successful in retaining.

The feature set involves:

- Built-up area (m^2)
- NDVI
- MNDWI
- Elevation (m)
- ESA Land Use/Land Cover classes (one-hot encoded)

The target variable is Land Surface Temperature (LST) in $^\circ\text{C}$.

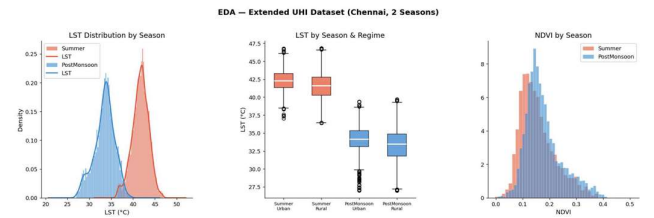


Fig. 2. LST Distribution & NDVI by Season

Outliers have been removed using the IQR method ($1.5\times$ threshold), eliminating a total of 517 samples (5.2%). The ESA Impervious class was excluded because of an almost-constant presence (95.7%), which resulted in limited predictive ability. The final dataset consists of 12 features.

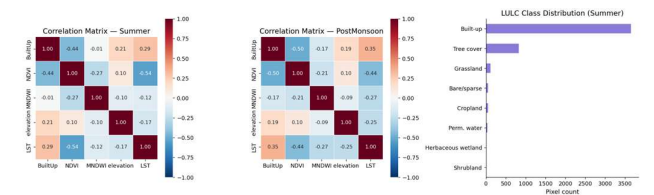


Fig. 3. Correlation Matrix and LULC Class Distribution

C. Descriptive Statistics

TABLE I
DATASET SUMMARY STATISTICS (N = 9,483)

Variable	Mean	Std Dev	Min	Max
BuiltUp (m ²)	3652.5	1744.3	0.0	9805.0
NDVI	0.166	0.069	-0.004	0.508
MNDWI	-0.144	0.044	-0.354	0.104
Elevation (m)	10.14	4.47	-8.0	55.0
LST (°C)	37.60	4.62	26.97	46.85

Seasonal separation is pronounced. Summer LST averages 41.70°C (SD = 2.4°C), while Post-Monsoon averages 33.47°C (SD = 3.4°C), indicating an 8.23°C difference. This drives the independent seasonal modelling.

D. Regime Definition

Pixels were classified into Urban and Rural regimes using a built-up threshold of 4,976 m² (dataset median). This gives us the classification:

- Summer: 25.2% Urban
- Post-Monsoon: 23.9% Urban

Statistical validation was performed using Mann–Whitney U and Levene’s tests ($p < 0.0001$), confirming distinct distributions.

The Urban Heat Island (UHI) effect is interpreted as:

$$UHI = LST_{\{urban\}} - LST_{\{rural\}UHI}$$

The Observed UHI values:

- Summer: 0.80°C
- Post-Monsoon: 0.86°C

E. Spatial Cross-Validation

To address spatial autocorrelation, block-based spatial cross-validation (5-fold) was employed instead of random splitting. Sequential partitioning was used to approximate geographic separation, reducing spatial leakage and ensuring more realistic generalisation.

F. Model Training

Models were trained across six configurations (Season × Regime: Global, Urban, Rural), using:

- Linear Regression (LR)
- Random Forest (RF)

Total models: 12

Linear Regression:

$$LST = \beta_0 + \sum \beta_j X_j + \epsilon$$

Random Forest:

$$L\hat{S}T = \frac{1}{T} \sum_{t=1}^T f_t(X)$$

Features were standardised for LR. RF models used default parameters with a fixed random state.

G. SHAP-Based Interpretation

TreeSHAP was applied to regime-specific RF models. Feature importance was computed using mean absolute SHAP values.

Seasonal stability was evaluated using:

$$\Delta = |SHAP_{\{summer\}} - SHAP_{\{post-monsoon\}}|$$

Features with $\Delta < 0.05$ were considered stable.

IV. ANALYSIS AND RESULTS

A. Model Performance

TABLE II
MODEL PERFORMANCE (SPATIAL CROSS-VALIDATION)

Season	Regime	Model	R ²	RMSE (°C)
Summer	Global	LR	0.420	1.478
Summer	Global	RF	0.447	1.443
Summer	Urban	LR	0.318	1.360
Summer	Urban	RF	0.289	1.389
Summer	Rural	LR	0.425	1.513
Summer	Rural	RF	0.448	1.482
Post-Monsoon	Global	LR	0.413	1.810
Post-Monsoon	Global	RF	0.487	1.692
Post-Monsoon	Urban	LR	0.372	1.681
Post-Monsoon	Urban	RF	0.374	1.678
Post-Monsoon	Rural	LR	0.419	1.837
Post-Monsoon	Rural	RF	0.498	1.708

RF generally outperforms LR, with the strongest performance in Post-Monsoon Rural ($R^2 = 0.498$).

Urban models consistently underperform relative to rural ones.

Wilcoxon tests show no significant difference between global and regime-specific model errors ($p > 0.05$), indicating comparable predictive accuracy.

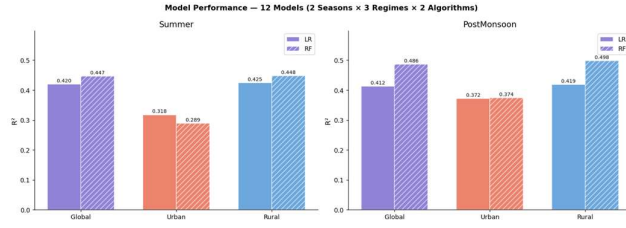


Fig. 4. Model Performance by Season

B. Feature Importance Trends

NDVI dominates in Summer across both regimes (importance > 0.5). Built-up density shows higher importance within the urban regime.

Post-Monsoon shifts are more pronounced:

- Urban: MNDWI increases significantly
- Rural: NDVI decreases, elevation increases

These shifts are consistent across both Gini importance and SHAP values.

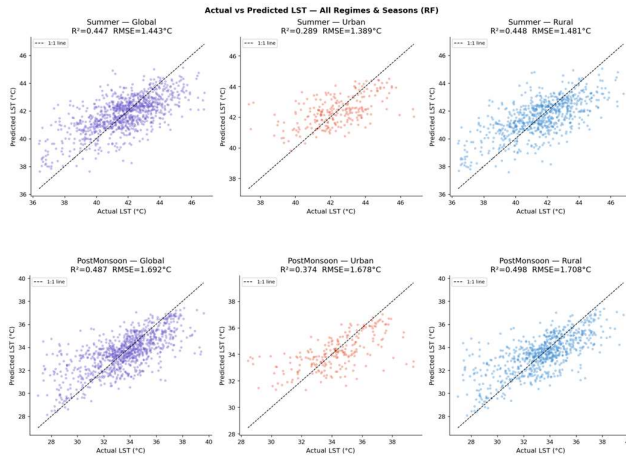


Fig. 5. Actual vs Predicted LST by Regimes and Seasons

C. SHAP Feature Attribution

TABLE III
MEAN |SHAP| VALUES

Feature	Summer Urban	Summer Rural	PM Urban	PM Rural
BuiltUp	0.176	0.109	0.105	0.276
NDVI	0.659	0.863	0.734	0.520
MNDWI	0.254	0.327	0.622	0.395
Elevation	0.114	0.289	0.263	0.651

NDVI seems to remain the dominant predictor across most settings.

However:

- MNDWI becomes critical in Post-Monsoon Urban
- Elevation becomes dominant in Post-Monsoon Rural

V. CONCLUSION

This study demonstrated that surface thermal dynamics in Chennai cannot be effectively captured using a single global predictive model. By separating pixels into urban and rural regimes and then training independent Random Forest models with spatial block cross-validation, applying SHAP-based analysis revealed that feature importance is structurally different across regimes and seasons by a substantial amount. NDVI dominates urban summer thermal prediction; MNDWI abruptly rises in the urban post-monsoon period; elevation turns out to become the dominant feature in the rural post-monsoon conditions. The Wilcoxon test suggests that regime-specific models do not significantly outperform global models on average error. Regardless, it does not contradict our findings. Instead it reinforces that the value of stratification lies in understanding the system rather than improving accuracy. For cities like Chennai, where heat distribution is uneven across a complex urban-rural landscape, models that study the mechanisms behind the heat are more useful for planning than models that simply minimise average prediction error.

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